REFONTE INTERNATIONAL TRAINING AND INTERNSHIP PROGRAM

Astrodynamics Specialist

SESSION 12

Capstone Build

Design, Optimization, Simulation and Verification

Prepared for	Cao Khanh Tan
Module	12, the capstone build and final deliverable
Builds on	Session 11 (mission definition) and the methods of Sessions 1 to 10
Deliverable	Completed capstone: design, verified results, report and defence
Effort	About 2 weeks at 12 to 15 hours

SESSION 12

Capstone Build: Design, Optimization, Simulation and Verification

With the mission defined in Session 11, this final module is the build: turning requirements into a design, refining it to the best you can, confirming it in simulation, and proving it meets every requirement. It is where the whole programme converges, the methods of Sessions 1 to 10 applied to a mission you specified yourself. The build is not a straight line but a loop: design, optimise, simulate, verify, and repeat until the design closes. This session walks that loop and carries one worked mission through it from end to end.

Learning objectives

- Turn a set of requirements into a baseline design by hand.
- Optimise the baseline into the best design that meets the requirements.
- Build a simulation that confirms and refines the design.
- Verify every requirement, cross-checking hand calculations against simulation.
- Run the design-simulate-verify loop to convergence and know when to stop.
- Assemble the completed capstone: results, report and defence.

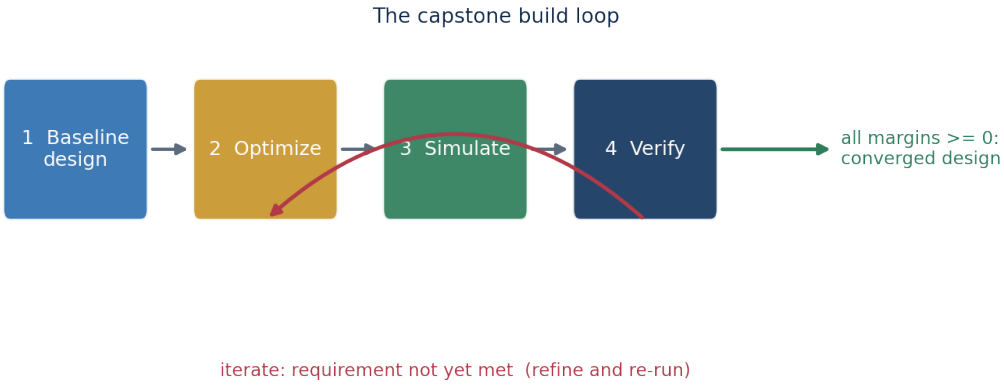


Figure 1. The capstone build loop. A baseline design is optimised, simulated and verified; if any requirement is unmet the design is refined and the loop repeats; when every margin is non-negative, the design has converged.

PART 1

From Requirements to a Baseline Design

Start by hand. Take each requirement and use the simplest relation that sizes it, the vis-viva equation for a transfer, the coverage geometry for a constellation, the Allen-Eggers formula for an entry, to produce a first baseline design. It will be rough, but it sets expectations and, crucially, it is the independent check that will later catch a simulation gone wrong. A baseline that is sized only inside a tool, with no hand estimate behind it, is the most common way capstones go quietly wrong.

Worked example, step 1: baseline (constellation track)
Requirement: mean revisit under 30 minutes and at least 90 percent coverage below 60 degrees latitude, seven-year life. From the Session 5 geometry, a satellite at 1200 km with a 10 degree mask sees a cap of central angle 24 degrees (about 4 percent of the Earth each). A first hand estimate says a few dozen satellites in a near-polar Walker pattern should do it; take a baseline of 1200 km, Walker 48/8/2 as the starting point.

PART 2

Optimise the Baseline

A baseline that works is not the end; the question from Session 10 is whether it is the best. Define the objective (fewest satellites, least delta-v), sweep the design variables, and find the smallest or cheapest design that still meets every requirement. Keep the trade visible so the choice is defensible.

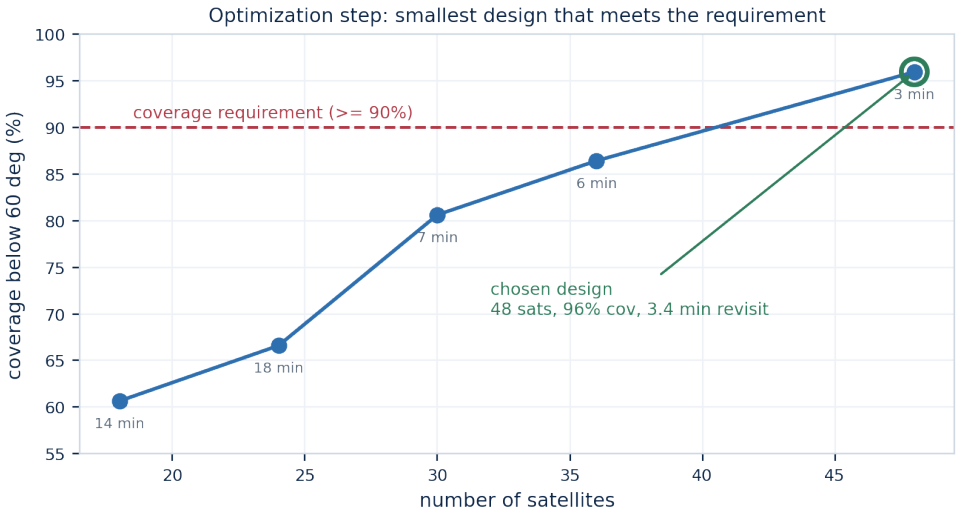


Figure 2. Worked example, step 2: optimisation. Sweeping the satellite count shows coverage climbing with size; the binding constraint is the 90 percent coverage line, and the smallest design that clears it (while easily meeting the revisit requirement) is the 48-satellite constellation. Fewer satellites save cost but fail coverage.

PART 3

Simulate

Now raise the fidelity. The hand design assumed circular orbits, a simple atmosphere, impulsive burns; the simulation adds what those assumptions left out, real perturbations, finite burns, actual geometry over time. Build the model in the tool that fits your track, STK or GMAT for trajectories and coverage, Python for custom analysis, and run it over a representative span (a full day for coverage, the whole transfer for a trajectory).

Worked example, step 3: simulation
The 48-satellite design is propagated over 24 hours and every grid point below 60 degrees is tested for access. The simulation returns a mean revisit of 3.4 minutes and 96 percent coverage, close to the hand expectation and comfortably inside both requirements. The orbital period (109 minutes) matches the Kepler hand value exactly, the first and simplest cross-check.

PART 4

Verify Against Every Requirement

Verification is the heart of the build: show, requirement by requirement, that the design meets what it must. Two checks run in parallel. First, hand and simulation must agree where both apply, to within a few percent; a large gap means one of them is wrong and must be found before trusting either:

$$\frac{|x_{\text{hand}} - x_{\text{sim}}|}{x_{\text{sim}}} < \varepsilon \quad \text{(a few percent)}$$

Second, every requirement must have a non-negative margin, the design is only verified when all of them do:

$$\text{verified} \iff \text{margin}_k \geq 0 \quad \forall k$$

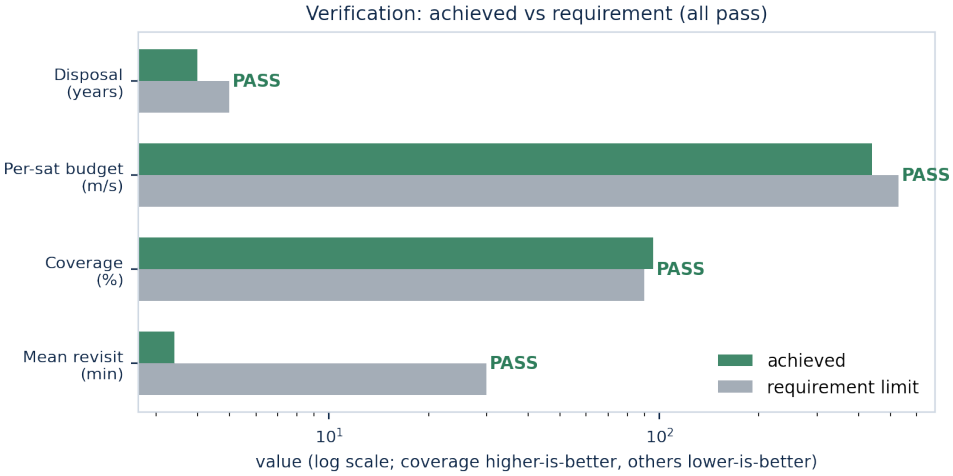


Figure 3. Worked example, step 4: verification. Each requirement is shown against its limit: mean revisit (3.4 vs 30 min), coverage (96 vs 90 percent), per-satellite delta-v (440 vs 530 m/s budget) and disposal time (4 vs 5 years). All margins are positive, so the design is verified.

The verification matrix, the table of requirement, method, predicted, simulated and result, is the single most important artefact of the capstone. It is what a reviewer reads first, and it is what the defence is built around.

PART 5

Iterate, Converge, and Know When to Stop

If any requirement fails, change the design and run the loop again. But convergence also means knowing when to stop: once every margin is comfortably positive, further optimisation is polishing, not engineering. A few common traps to avoid:

- **Unverified simulation:** trusting a tool output with no hand check behind it. Always keep the baseline as a sanity line.
- **Over-optimisation:** chasing the last one percent of a metric while adding fragility or complexity the requirement never asked for.
- **Scope creep:** adding capability beyond the mission definition. If it is not a requirement, it is not in the build.
- **Hidden assumptions:** a result that quietly depends on a circular orbit or a perfect atmosphere. State every assumption and test the ones that matter.

What the completed build contains

A converged design with every requirement verified; a verification matrix; the simulation artefacts and plots; a trade study justifying the design; and a short limitations section. These become the mission report, and the report plus a defence is the capstone.

The Build Workflow in Practice

The loop is the same whatever the track and whatever the tool: hand first, then optimise, then simulate, then verify against the requirement.

```
# the capstone build loop, in pseudocode
baseline = size_by_hand(requirements)          # Part 1: vis-viva / coverage / Allen-Eggers
design   = optimize(baseline, objective, requirements)   # Part 2: Session 10 methods

while True:
    results = simulate(design)                 # Part 3: STK / GMAT / Python, full fidelity
    margins = verify(results, requirements)    # Part 4: achieved vs required, hand vs sim
    if all(m >= 0 for m in margins):
        break                                  # converged: every requirement met
    design = refine(design, margins)           # Part 5: fix the failing requirement, repeat
```

Production tools, one last time

You have the full kit: Python (numpy, scipy, poliastro) for hand checks and custom analysis, STK (Coverage, Astrogator, Lifetime, Navigation) and GMAT (targeting, optimisation, high-fidelity propagation) for the simulation. Use whichever fits the track; just keep a hand calculation beside every headline number the tool produces.

PRACTICE

Build Your Capstone (about 2 weeks)

Execute the build for the mission you defined in Session 11, then write it up and defend it. Work the loop in order and keep the verification matrix current throughout.

Deliverable: the completed capstone

- **Baseline design (3 to 4 h).** Size a first design by hand from your Session 11 requirements. Show every calculation and record the predicted value for each requirement.
- **Optimisation (3 to 4 h).** Define your objective, sweep the design variables, and select the best design that meets all requirements. Include the trade study and justify the choice.
- **Simulation (4 to 6 h).** Build the model in STK, GMAT or Python and run it over a representative span. Export the key plots (trajectory, coverage or figure-of-merit maps, delta-v tables).
- **Verification (3 to 4 h).** Complete the verification matrix: requirement, method, predicted, simulated, margin, result. Confirm hand and simulation agree to within a few percent, and that every margin is non-negative. Iterate if not.
- **Report and defence (4 to 6 h).** Assemble the mission report (definition, design, optimisation, simulation, verification, limitations) and prepare the 15 to 20 minute oral defence.

Verification matrix (the template to fill in)

Requirement	Method	Predicted (hand)	Simulated	Result
Mean revisit < 30 min	simulation	~ (your estimate)	3.4 min	PASS
Coverage >= 90 percent	simulation	~ (your estimate)	96 percent	PASS
Per-sat delta-v <= 530 m/s	analysis	440 m/s	n/a	PASS
Disposal within 5 years	analysis	deorbit feasible	4.0 years	PASS

The finish line: your defence

When the matrix is all green and the report is written, you present and defend it. That completes the Astrodynamics Specialist programme: you will have taken a mission from a blank page, through definition, design, optimisation, simulation and verification, to a defended result, using the same methods and tools the space industry uses.

REFERENCE

Build Checklist and Tools

Work the loop in this order, and do not skip the hand check:

- **Baseline:** size every requirement by hand; record the predicted value.
- **Optimise:** objective chosen, design variables swept, smallest/cheapest feasible design selected, trade shown.
- **Simulate:** full-fidelity model over a representative span; key plots exported.
- **Verify:** hand vs simulation agree to a few percent; every requirement has a non-negative margin.
- **Iterate:** fix the failing requirement and re-run; stop when all margins are comfortably positive.
- **Document:** verification matrix, trade study, plots, limitations, into the report.

Quick formulas

Hand vs simulation agreement	$\frac{\lvert x_{\text{hand}} - x_{\text{sim}} \rvert}{x_{\text{sim}}} < \varepsilon$ (a few percent)
Design verified	$\text{verified} \Leftrightarrow \text{margin}_k \geq 0 \ \ \forall k$

Key glossary

- **Baseline design:** the first, hand-sized design that meets the requirements roughly.
- **Verification matrix:** the table proving each requirement is met; the capstone centrepiece.
- **Margin:** the headroom between achieved and required; verified means all margins non-negative.
- **Fidelity:** how fully a model captures reality; hand low, simulation high, and they should agree.
- **Convergence:** the point where the design meets every requirement and iteration can stop.

Going further

For the build and write-up, the programme references carry through: Curtis and Vallado for the astrodynamics, Wertz and Larson for mission-level design and verification. These are real, widely used texts; please confirm editions yourself, as I cannot verify those details from here.

Programme complete

Session 12 closes the taught and capstone sequence of the Astrodynamics Specialist programme. From the two-body problem to a defended mission, the full arc is yours. Build something you would be proud to present.